A3 Solutions: First algebra

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Try every problem properly before reading its solution.

Practice

1.

(a) $4a - 2a = 2a$ and $3b - 5b = -2b$: $2a - 2b$.

(b) $5 \cdot (-2) \cdot x \cdot x = -10x^2$.

(c) $ba = ab$, so $3ab - 5ab = -2ab$: $-2ab + 2a$. (ab and a are not like terms.)

(d) $6x^2 - 2x - 7$.

(e) $(-3) \cdot 4 \cdot (-1) \cdot p \cdot p \cdot q = 12p^2q$.

(f) $a \cdot a \cdot a = a^3$, so $a^3 + 2a^3 = 3a^3$.

2.

(a) $12x - 8$

(b) $-6 + 15a$ (both terms change sign: $-3 \cdot 2 = -6$ and $-3 \cdot (-5a) = 15a$)

(c) $7 - 4 + 3x = 3 + 3x$

(d) $2x + 10 - 3x + 3 = -x + 13$ (note: $-3 \cdot (-1) = +3$)

(e) $2x^2 - 3x - x^2 = x^2 - 3x$

(f) $x^2 + 7x + 2x + 14 = x^2 + 9x + 14$

(g) $2a^2 + 8a - a - 4 = 2a^2 + 7a - 4$

(h) $(x - 3)(x - 3) = x^2 - 3x - 3x + 9 = x^2 - 6x + 9$

3.

(a) $4x = 24$, $x = 6$. Check: $24 + 7 = 31$ ✓.

(b) Add $2x$: $7 = 5x - 8$. Add 8: $15 = 5x$, so $x = 3$. Check: $7 - 6 = 1$ and $9 - 8 = 1$ ✓.

(c) $4x = 14$, $x = 3\frac{1}{2}$. Check: $21 - 5 = 16$ and $7 + 9 = 16$ ✓.

(d) $4x + 4 - 2x + 6 = 20$, so $2x + 10 = 20$, $x = 5$. Check: $4 \cdot 6 - 2 \cdot 2 = 24 - 4 = 20$ ✓.

(e) $\frac{x}{4} = 3$, $x = 12$. Check: $3 + 2 = 5$ ✓.

(f) Multiply both sides by 6: $3(x + 3) - 2(x - 1) = 18$, so $3x + 9 - 2x + 2 = 18$, $x + 11 = 18$, $x = 7$. Check: $\frac{10}{2} - \frac{6}{3} = 5 - 2 = 3$ ✓.

4.

(a) $K = 25 + 40t$.

(b)

t (hours)	0	0.5	1	2	3
K (€)	25	45	65	105	145

The graph is a straight line through $(0, 25)$ rising 40 per hour.

(c) $25 + 40 \cdot 2.5 = 25 + 100 =$ **€125**.

(d) $25 + 40t = 165$, so $40t = 140$, $t = 3.5$: **3½ hours**.

5.

(a) Difference 4, first term 7: $u_n = 7 + 4(n-1) = 4n + 3$.

(b) $u_{50} = 4 \cdot 50 + 3 = 203$.

(c) $4n + 3 = 403$ gives $n = 100$: **yes**, term 100. $4n + 3 = 500$ gives $4n = 497$, $n = 124.25$, not a whole number: **no**. (Also: every term leaves remainder 3 on division by 4; 500 leaves 0.)

6.

(a) $2n + 2$. Picture: each table has one person on its top side and one on its bottom side ($2n$ people), plus one person at each end of the row (2 more).

(b) $2n + 2 \geq 30$ gives $n \geq 14$. With exactly 14 tables: $2 \cdot 14 + 2 = 30$. **14 tables.**

7.

(a) $5x > 15$, $x > 3$. (Open dot at 3, arrow to the right.)

(b) $4 - 10 \leq 3x$, so $-6 \leq 3x$, $x \geq -2$. (Filled dot at -2, arrow to the right.)

(c) $2x - 2 \geq 5x + 4$, so $-6 \geq 3x$, $x \leq -2$. (Filled dot at -2, arrow to the left.)

(d) $\frac{x}{3} < 3$, $x < 9$. (Open dot at 9, arrow to the left.)

8.

(a) $(-3)^2 - 2 \cdot 2 = 9 - 4 = 5$

(b) $-(-3)^2 + (-3) \cdot 2 = -9 - 6 = -15$

(c) $(-3 + 2)^2 = (-1)^2 = 1$

(d) $3 \cdot (-3) - 2 \cdot (2 - (-3)) = -9 - 2 \cdot 5 = -19$

(e) $\dfrac{-3 - 2}{-3 + 2} = \dfrac{-5}{-1} = 5$

School-exam level

9.

(a) $K_A = 12 + 0.10m$ and $K_B = 20 + 0.05m$.

(b) $12 + 0.10m = 20 + 0.05m \Rightarrow 0.05m = 8 \Rightarrow m = 160$. At **160 minutes** both cost €28.

(c) Both graphs are straight lines; A starts lower (12) but is steeper (0.10 per minute), so the lines cross at $m = 160$. At $m = 250$: $K_A = 37$ and $K_B = 32.50$. **Plan B is cheaper.**

10.

Let a be Anna's age now. Her brother is $a-4$. In 5 years: $(a+5)+(a-4+5) = 40$, so $2a+6 = 40$, $a = 17$.

Check: now 17 and 13; in 5 years 22 and 18, sum 40 ✓. **Anna is 17 and her brother is 13.**

11.

Let the width be w cm; the length is $w + 5$. Perimeter: $2(w + w + 5) = 46$, so $4w + 10 = 46$, $w = 9$.

Width 9 cm, length 14 cm. Check: $2 \cdot (9 + 14) = 46$ ✓.

12.

Let the smallest be n: $n + (n + 1) + (n + 2) + (n + 3) = 154$, so $4n + 6 = 154$, $n = 37$. **37, 38, 39, 40.** Check: $37 + 38 + 39 + 40 = 154$ ✓.

13.

Let t be the time in hours. Together they cover 60 km: $18t + 22t = 60$, $40t = 60$, $t = 1.5$.

They meet after **1½ hours**, when the cyclist from A has ridden $18 \cdot 1.5 =$ **27 km** from A. (Check: the other has ridden $22 \cdot 1.5 = 33$ km, and $27 + 33 = 60$ ✓.)

14.

(a) $N = 7 \cdot 25 - 30 = 145$ chirps per minute.

(b) $N + 30 = 7T$, so $T = \dfrac{N + 30}{7}$.

(c) $T = \dfrac{110 + 30}{7} = \dfrac{140}{7} = 20$ °C.

(d) $N = 0$ when $T = \frac{30}{7} \approx 4.3$ °C. Below about **4.3 °C** the formula gives a negative number of chirps, which means the crickets are silent. (In reality such a rule of thumb only works in a limited range of temperatures.)

15.

(a) Pairs from both ends add to 201; there are 200 terms. $\dfrac{200 \cdot 201}{2} = 20\,100$.

(b) $2 + 4 + \cdots + 200 = 2(1 + 2 + \cdots + 100) = 2 \cdot 5050 = 10\,100$.

(c) The terms are $4n - 1$ for $n = 1, \ldots, 50$ (since $4 \cdot 50 - 1 = 199$): 50 terms. Writing it twice, forwards and backwards, each pair adds to $3 + 199 = 202$. So the sum is $\dfrac{50 \cdot 202}{2} = 5050$.

16.

Let x be the number of 20-cent coins; then there are $25 - x$ fifty-cent coins. In cents: $20x + 50(25 - x) = 860$, so $1250 - 30x = 860$, $30x = 390$, $x = 13$.

13 coins of 20 cents and 12 of 50 cents. Check: $260 + 600 = 860$ ✓.

17.

Let n be the number of pupils. Money in: $25n$. Costs: $450 + 12n$. We need $25n \geq 450 + 12n$, so $13n \geq 450$, $n \geq 34.6\ldots$ The number of pupils is whole, so **at least 35 pupils**.

Check: 35 pupils pay €875 and the costs are $450+420 = 870$ ✓; 34 pupils pay €850 and the costs are €858 ✗.

18.

(a) Slope $a = \dfrac{20-11}{5-2} = 3$. Then $11 = 3 \cdot 2 + b$ gives $b = 5$: $y = 3x + 5$.

(b) $3 \cdot 10 + 5 = 35 \neq 36$: **no**.

(c) $3x + 5 = 0$ gives $x = -\frac{5}{3}$.

19.

(a) $3x - 6 + 6 = 3x$, so $3x = 3x$, and then $0 = 0$. True for **every** x: both sides are the same expression.

(b) $4x - 2x - 2 = 2x + 5$, so $2x - 2 = 2x + 5$, and then $-2 = 5$. Never true: **no solution**.

(c) The right side is $5x + 10$. The equation $5x + c = 5x + 10$ becomes $c = 10$. If $c = 10$, every x is a solution; if $c \neq 10$, there is no solution.

Challenge

20.

(a) $1, 4, 9, 16$: the squares. Guess: $1 + 3 + \cdots + (2n-1) = n^2$.

(b) Write the sum S twice, once backwards:

$$\begin{array}{rcccccccc} S & = & 1 & + & 3 & + & \ldots & + & (2n-1) \\ S & = & (2n-1) & + & (2n-3) & + & \ldots & + & 1 \end{array}$$

There are n terms, and each column adds to $2n$. So $2S = n \cdot 2n$, $S = n^2$. □

(c) Build an n by n square of dots in L-shaped layers starting from a corner: the corner dot (1), then an L of 3 dots making a 2 by 2 square, then an L of 5 making 3 by 3, and so on. Going from a $(k-1)$ by $(k-1)$ square to a k by k square adds $k + (k-1) = 2k - 1$ dots. So $1 + 3 + 5 + \cdots + (2n-1)$ dots fill the n by n square: n^2. □

21.

$2n + 7 = 2(n+1) + 5$, so

$$\frac{2n+7}{n+1} = 2 + \frac{5}{n+1}.$$

This is an integer exactly when $n+1$ divides 5, that is $n+1 \in \{1, -1, 5, -5\}$. So $n \in \{0, -2, 4, -6\}$.

Check: $n = 0$: 7; $n = -2$: $\frac{3}{-1} = -3$; $n = 4$: $\frac{15}{5} = 3$; $n = -6$: $\frac{-5}{-5} = 1$. All integers ✓.

22.

(a) Call the number x. The steps give $4x$, $4x+12$, $\frac{4x+12}{2} = 2x+6$, and $2x+6-2x = 6$. Always 6, whatever x was. □

(b) $(x+3)(x-3)+9 = x^2-9+9 = x^2$. □

(c) Many answers. Example: "Think of a number. Multiply by 3. Add 15. Divide by 3. Subtract the number you started with." Proof: $\frac{3x+15}{3} - x = x+5-x = 5$. □ (The trick: arrange the steps so the x is cancelled at the end.)

23.

(a) Split a square of side $a+b$ by cutting each side into pieces a and b. You get a square a by a, a square b by b, and **two** rectangles a by b. Total area: $a^2+2ab+b^2$.

(b) $103^2 = (100+3)^2 = 10\,000+600+9 = 10\,609$. $98^2 = (100-2)^2 = 10\,000-400+4 = 9604$.

(c) $97 \cdot 103 = (100-3)(100+3) = 10\,000-9 = 9991$.

(d) $a = b = 1$: $(1+1)^2 = 4$ but $1^2+1^2 = 2$. In the picture, he has forgotten the two rectangles ab.

24.

We need whole numbers $x, y \geq 0$ with $3x+5y = 41$. Try each y from 0 to 8 (since $5 \cdot 9 > 41$) and see whether $41-5y$ is divisible by 3:

y	0	1	2	3	4	5	6	7	8
$41-5y$	41	36	31	26	21	16	11	6	1
divisible by 3?	no	**yes**	no	no	**yes**	no	no	**yes**	no

So **(12 pens, 1 notebook), (7 pens, 4 notebooks), (2 pens, 7 notebooks)**.

Notice the pattern: the solutions differ by 5 pens and 3 notebooks, because swapping 5 pens (€15) for 3 notebooks (€15) keeps the total the same.

25.

(a) Write the sum S forwards and backwards. The last term is $a+(n-1)d$. Each column adds to $a+(a+(n-1)d) = 2a+(n-1)d$, and there are n columns:

$$2S = n\,(2a+(n-1)d), \qquad S = \frac{n\,(2a+(n-1)d)}{2}. \quad \square$$

(b) $2027 = 1013+1014$. (Any odd number $2m+1 \geq 3$ is $m+(m+1)$.)

(c) Suppose $64 = m+(m+1)+\cdots+(m+k-1)$ with $k \geq 2$ and $m \geq 1$. By (a) with $a = m$, $d = 1$: $64 = \frac{k(2m+k-1)}{2}$, so $k(2m+k-1) = 128$.

The two factors k and $2m+k-1$ differ by $2m-1$, which is odd, so one of them is odd and the other even. The odd one divides $128 = 2^7$, so it must be 1. But $k \geq 2$ and $2m+k-1 \geq 2+2-1 = 3$, so neither factor is 1. Contradiction. So 64 is not such a sum. □

(The same argument shows that **no power of 2** is a sum of two or more consecutive positive whole numbers, and in fact every other positive number is. Try to prove that!)

26.

The number is $100a + 10b + c$ and its reverse is $100c + 10b + a$. The difference is

$$(100a + 10b + c) - (100c + 10b + a) = 99a - 99c = 99(a - c),$$

which is divisible by 99. □

The difference is 0 exactly when $a = c$: the number is a palindrome, like 353 or 707.

27.

(a) There: $60 \div 30 = 2$ hours. Back: $60 \div 60 = 1$ hour. Total 120 km in 3 hours: **40 km/h**. The average is not 45 because you spend *more time* at the slow speed.

(b) Time there $\frac{d}{u}$, back $\frac{d}{w}$. Average speed:

$$\frac{2d}{\frac{d}{u} + \frac{d}{w}} = \frac{2d}{\frac{dw+du}{uw}} = \frac{2d \cdot uw}{d(u+w)} = \frac{2uw}{u+w}. \quad \square$$

(This is called the *harmonic mean* of u and w.) Check with (a): $\frac{2 \cdot 30 \cdot 60}{90} = 40$ ✓.

(c) An average of 60 km/h over 120 km means 2 hours in total. But the way there already took 2 hours. So it's **impossible**, however fast you drive back. Algebraically: $\frac{2 \cdot 30w}{30+w} = 60$ gives $60w = 1800 + 60w$, so $0 = 1800$: no solution.

28.

(a) Let the numbers be x (larger) and y. Adding "$x + y = 20$" and "$x - y = 6$": $2x = 26$, $x = 13$, and then $y = 7$. Check: $13 + 7 = 20$, $13 - 7 = 6$ ✓.

(b) In the same way $2x = S + D$ and $2y = S - D$, so

$$x = \frac{S + D}{2}, \qquad y = \frac{S - D}{2}.$$

(The larger number is the average plus half the difference.)

29.

(a) $3x - 6 - 2x - 2 > x - 10$, so $x - 8 > x - 10$, so $-8 > -10$. That's always true, whatever x is: **every number is a solution**.

(b) For example $x + 1 < x$. Subtracting x gives $1 < 0$, which is false. No number is 1 more than itself and smaller than itself, so there are **no solutions**.